Congyan (Cruise) Song

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Research Interests

Formal verification, broadly — with a focus on:

- Formalization of mathematics
- Theorem proving & automated reasoning
- Verification of parsers & compilers
- Autoformalization & neurosymbolic AI

Education

Georgia Institute of Technology, Atlanta, GA — Aug. 2024 – Present

Ph.D. in Computer Science. Advisor: Dr. Vijay Ganesh.

- Selected coursework: SAT and SMT Solvers, Software Analysis and Test, Programming Languages

University of Michigan, Ann Arbor, MI — Jul. 2021 – May 2024

B.S. in Mathematics and Computer Science.

Publications

1. **C. Song**, H. Duong, T. Nguyen, V. Ganesh. "LeanVIPR: A Formally Verified Certificate Checker for Exact Mixed-Integer Programming." NeurIPS Workshop on AI for Verifiable Coding, 2026.
2. P. Jha, **C. Song**, Z. Li, Y. Sejpal. "VeriPIE: Checkable Query Plans and Evidence-Guided Selection for Text-to-SQL." Preprint.
3. **C. Song**, Z. Li, S. Binder, V. Ganesh. "Just Draw It: Widgets for Generating and Visualizing Graph Structures in Lean 4." Workshop on Human Aspects of Types and Reasoning Assistants, 2026.
4. P. Jana, K. Kale, E. Tanrıverdi, **C. Song**, S. Vishwanath, V. Ganesh. "ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings." ICLR 2026.
5. D. Narváez, **C. Song**, N. Zhang. "Formalizing Finite Ramsey Theory in Lean 4." Conference on Intelligent Computer Mathematics (CICM), 2024.

Research Experience

Georgia Institute of Technology, Atlanta, GA — Aug. 2024 – Present

- Explored combinations of Lean 4 with SAT solvers and computer algebra systems for combinatorial problems.
- Contributed to autoformalization efforts, building systems that translate natural-language theorems and proofs into Lean.

The Polymath Jr. Program (Online) [code] — Jun. 2022 – Aug. 2024

- Built proficiency with interactive theorem proving in Lean 3 and Lean 4.
- Formalized graph theory problems using interactive and automated theorem proving with SAT solvers.
- Developed a Lean widget that interactively visualizes the graphs used in proofs.

University of Michigan, Ann Arbor, MI — Sep. 2022 – May 2024

- Studied formal verification of iterative methods and floating-point arithmetic in Rocq.
- Worked on proof minimization and refactoring by developing Ltac tactics in Rocq.
- Thesis: *An Attempt to Verify the Correctness, Accuracy and Convergence of the Block Jacobi Method.*

Industry Experience

Applied Scientist Intern, **Amazon Inc.**, Santa Clara, CA [project] — May 2026 – Present

- Verified the extension parser of the Cedar policy language in Lean.
- Built a grammar-to-spec compiler and a verified parser generator for format parsers in Lean.

Functional Software Intern, **Imandra Inc.**, Austin, TX [code] — May 2025 – Aug. 2025

- Extended Imandra-Geo, a solver for non-linear real arithmetic, to emit proof certificates consumable by Lean.
- Developed custom Lean tactics via metaprogramming to normalize non-linear real arithmetic expressions and translate Lean syntax into Imandra-Geo input format, working in OCaml, ACL2, and Imandra.

Teaching

- **Teaching Assistant**, CS 4510 Automata and Complexity — Georgia Tech.
- **Teaching Assistant**, Math 215 Multivariable & Vector Calculus — University of Michigan.

Skills

Programming: Lean 4, Python, C++, OCaml. **Languages:** Mandarin (native), English (proficient).